Zehong Liao

Nuclear Reaction Theory / Low-Energy Heavy-Ion Reactions

Email liaozh26@mail2.sysu.edu.cn **Website** murry-liao.github.io **GitHub** murry-liao

Academic Overview

PhD in Physics, Sun Yat-sen University (expected July 2026). Works on the theory of low-energy heavy-ion reactions, with a focus on multinucleon transfer, isotope-production mechanisms, and sub-barrier fusion. Author of 13 publications (6 as first author); research visitor at Kyoto University, Japan.

Education & Research Experience

Sep 2021 – Jul 2026 — **Sun Yat-sen University**, China · PhD in Physics
Sino-French Institute of Nuclear Engineering and Technology; Supervisor: Prof. Long Zhu.

Apr 2025 – Dec 2025 — **Kyoto University**, Japan · Research Visitor
Department of Physics, Graduate School of Science; Host: Prof. Kouichi Hagino.

Sep 2017 – Jun 2021 — **North China Electric Power University**, China · B.Sc.
Nuclear Science and Technology.

Research Directions

- **Multinucleon Transfer Reactions (MNT)** Nucleon exchange, charge equilibration, shell effects, and their influence on the production cross sections and angular distributions of neutron-rich $N = 126$ isotones.
- **Low-Energy Heavy-Ion Reaction Dynamics** Quasielastic scattering, transfer, and fusion studied via the dinuclear system model, master equations, and semiclassical dynamics; developing a unified Coupled Master–Langevin model.
- **Nuclear Reaction Computation Methods** Coupled-channels theory, reduced-basis and emulator methods, for sub-barrier fusion cross-section calculations and constraining nuclear deformation parameters.

Talks & Academic meeting (Selected)

2026.10 — **A unified description of low-energy fusion and multinucleon-transfer reactions within the Coupled Master–Langevin model** · Seminar, Department of Physics, Kyoto University

2025.12 — **Determination of nuclear deformations with an emulator for sub-barrier fusion reactions** · 10th Kansai Nuclear Multi-Body Seminar, Osaka Metropolitan University, Japan

2024.11 — **Angular distribution in the multinucleon transfer reaction** · 3rd Symposium on Nuclear Physics in the Guangdong–Hong Kong–Macao Greater Bay Area, Shenzhen, China

2024.10 — **Angular distribution in the multinucleon transfer reaction** · 2nd Multinucleon Transfer Reactions Symposium, Changsha, China

Open Source & Research Software

Coupled-channels model & emulator: Fortran 90 / Python solver for heavy-ion fusion coupled-channels calculations and reduced-basis emulator development; includes a rewrite of CCFULL and multiple numerical solution methods. Project repository

Nuclear potential & mass calculations: Fortran program for Coulomb-barrier and potential-energy-surface description; other in-house codes include the dinuclear system model, FRLDM, and TCSM. Project repository

Representative Publications

13 publications in total; 6 first-author papers and 2 representative collaborative works are listed below.

01 **Determination of nuclear deformations with an emulator for sub-barrier fusion reactions** [First Author]
Zehong Liao, K. Hagino, Long Zhu, S. Yoshida, K. Uzawa. Phys. Rev. C **113**, 044611 (2026). · DOI

02 **Mechanism of the quasielastic scattering based on the dinuclear system concept** [First Author · Letter]
Zehong Liao, Yu Yang, Zepeng Gao, Jun Su, Long Zhu. Phys. Rev. C **112**, L051601 (2025). · DOI

03 **Quantifying angular distributions in multinucleon transfer reactions with a semiclassical method** [First Author]
Zehong Liao et al. Phys. Rev. C **111**, 024605 (2025). · DOI

04 **Shell effects on the drift and fluctuation in multinucleon transfer reactions** [First Author]
Zehong Liao et al. Phys. Rev. C **109**, 054612 (2024). · DOI

05 **Optimal detection angles for producing $N = 126$ neutron-rich isotones in multinucleon transfer reactions** [First Author · Letter]
Zehong Liao et al. Phys. Rev. Research **5**, L022021 (2023). · DOI

06 **Dynamics of charge equilibration and effects on producing neutron-rich isotopes around $N = 126$ in multinucleon transfer reactions** [First Author]
Zehong Liao et al. Phys. Rev. C **107**, 014614 (2023). · DOI

07 **Correlation of multiple physical quantities in multinucleon transfer reactions: insight into production of isotopes near the shell closure $N = 152$** [Co-author · Editors' Suggestion]
Zimeng Shen, Zehong Liao, Long Zhu. Phys. Rev. C **113**, L021601 (2026). · DOI

08 **Emulating multichannel scattering based on eigenvector continuation in the discrete-basis formalism** [Co-author]
K. Hagino, Zehong Liao, S. Yoshida, M. Kimura, K. Uzawa. Phys. Rev. C **112**, 024618 (2025). · DOI